

# Adaptive Ant Colony Metaheuristic Optimization for Nonlinear Programming Problems: A Hybrid AACO–SQP Numerical Study

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**Abstract**—Ant Colony Optimization (ACO) is a population-based metaheuristic inspired by collective foraging behavior and has achieved notable success in discrete combinatorial optimization. Continuous-domain extensions are attractive for nonlinear programming, but practical use is often hindered by parameter sensitivity, discretization effects, and limited final precision. This paper implements an Adaptive Ant Colony Optimization (AACO) framework for continuous optimization and compares it to Sequential Quadratic Programming (SQP) and L-BFGS-B. Motivated by limitations reported in prior continuous-ACO literature, we propose a Hybrid AACO–SQP pipeline that couples global exploration (AACO) with local refinement (SQP). Experiments on standard 5D benchmark functions (Rosenbrock and Rastrigin) over 30 independent runs evaluate solution quality, robustness, and computational cost. Results demonstrate that the hybrid approach achieves superior accuracy (mean objective  $2.73 \times 10^{-7}$  on Rosenbrock) while maintaining robustness on multimodal landscapes (Rastrigin:  $1.23 \pm 0.43$ ).

**Index Terms**—Ant Colony Optimization, Adaptive Metaheuristics, Sequential Quadratic Programming, Hybrid Optimization, Nonlinear Programming, Benchmark Optimization.

## I. INTRODUCTION

Optimization is fundamental in engineering and scientific computing. Classical numerical methods such as SQP are efficient for smooth problems and can achieve fast local convergence under standard regularity assumptions, but they are sensitive to initialization and may converge to undesired local minima in non-convex landscapes [5].

Metaheuristics emphasize stochastic global exploration and can reduce dependence on initial guesses. ACO, introduced by Dorigo and Stützle, is a prominent metaheuristic originally designed for discrete problems [1]. Extending ACO to continuous domains is an active area; Socha and Dorigo proposed a foundational continuous ACO formulation that replaces discrete path construction with continuous sampling [2].

### A. Contributions

The contributions of this course project are:

- An implementable AACO procedure for continuous optimization inspired by continuous-ACO literature with adaptive parameter scheduling.
- A two-phase Hybrid AACO–SQP pipeline combining global search with local refinement.

- A reproducible evaluation protocol (30 independent runs per method, convergence curves, boxplots, CSV exports) aligned with Math 303 objectives and best practices in computational optimization research.

## II. OPTIMIZATION BACKGROUND

### A. Nonlinear Programming

A general constrained optimization problem is:

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad g(x) \leq 0, \quad h(x) = 0. \quad (1)$$

Nonlinear programs may be non-convex and contain multiple local minima, making initialization and global search strategies critical [6].

### B. Sequential Quadratic Programming

SQP solves constrained nonlinear optimization by iteratively approximating the problem with quadratic programming subproblems using gradient and Hessian (or quasi-Newton approximation) information [5]. SQP exhibits superlinear local convergence near an optimum but may perform poorly if started far from a good basin of attraction.

### C. Continuous ACO

Continuous ACO variants replace discrete construction with probabilistic sampling mechanisms over a continuous domain [2]. In this work, we discretize each dimension into bins and maintain a pheromone matrix that guides sampling via attractiveness-weighted probabilities. Performance depends on parameter selection and the representation/sampling model, motivating adaptive parameter control strategies [3].

## III. RELATED WORK AND RESEARCH GAPS

Prior work shows continuous-domain ACO is feasible but faces practical implementation challenges [2]. Survey and methodological discussions highlight that parameter control is a key issue in ACO implementations [3].

The main practical gaps motivating this project are:

- **Parameter sensitivity:** Manual tuning of pheromone-related parameters can reduce robustness across instances [3] [2].

- **Limited hybridization in reproducible coursework settings:** Hybrid metaheuristic–numerical ideas exist [4], but course-level implementations comparing ACO-style global search with SQP-style local refinement are limited.
- **Precision plateaus:** Discretization/sampling choices can cause slow final improvements in continuous ACO, motivating a local solver refinement stage [2].

#### IV. METHODOLOGY

##### A. Benchmark Problems

Two standard test functions in 5 dimensions are used:

**Rosenbrock (5D):**

$$f_{\text{Rosenbrock}}(x) = \sum_{i=1}^4 [100(x_{i+1} - x_i^2)^2 + (1 - x_i)^2], \quad (2)$$

with bounds  $x_i \in [-3, 3]$  and global minimum at  $(1, 1, 1, 1, 1)$  where  $f = 0$ .

**Rastrigin (5D):**

$$f_{\text{Rastrigin}}(x) = 50 + \sum_{i=1}^5 [x_i^2 - 10 \cos(2\pi x_i)], \quad (3)$$

with bounds  $x_i \in [-5.12, 5.12]$  and global minimum at  $(0, 0, 0, 0, 0)$  where  $f = 0$ .

These benchmarks are widely used to test local-vs-global behavior: Rosenbrock features a narrow parabolic valley (testing convergence precision), and Rastrigin is highly multimodal (testing global exploration).

##### B. Adaptive Ant Colony Optimization (AACO)

Each ant represents a candidate solution in the continuous search space. AACO discretizes each dimension into 60 bins and maintains a pheromone matrix  $\tau_{d,j}$  for dimension  $d$  and bin  $j$ . At iteration  $t$ , parameters  $(\alpha, \beta, \rho)$  are linearly interpolated between initial and final values to balance exploration (early) and exploitation (late):

$$\alpha(t) = \alpha_0 + \frac{t}{T-1}(\alpha_1 - \alpha_0), \quad (4)$$

$$\beta(t) = \beta_0 + \frac{t}{T-1}(\beta_1 - \beta_0), \quad (5)$$

$$\rho(t) = \rho_0 + \frac{t}{T-1}(\rho_1 - \rho_0). \quad (6)$$

Each ant samples dimension  $d$  from bin  $j$  with probability proportional to:

$$p_{d,j} \propto \tau_{d,j}^\alpha \cdot \eta_{d,j}^\beta, \quad (7)$$

where  $\eta_{d,j} = 1/(|g_d[j] - \text{mid}_d| + 0.1)$  is a heuristic attractiveness favoring bins near the domain center. After evaluation, the top 15% elite ants deposit pheromone inversely proportional to their objective values, and pheromones evaporate at rate  $\rho(t)$ .

##### C. Projected Gradient Descent (PGD)

A scratch implementation of projected gradient descent uses finite-difference gradients with step size  $\eta = 10^{-3}$  and projects solutions onto box constraints after each update. PGD runs for 250 iterations starting from the AACO solution.

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##### Algorithm 1 Adaptive Ant Colony Optimization (AACO)

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1: Input: objective  $f$ , bounds  $[l, u]$ , ants  $m = 40$ , iterations  $T = 150$ 
2: Initialize pheromone matrix  $\tau \leftarrow \mathbf{1}$ 
3: Initialize  $(\alpha_0, \beta_0, \rho_0) = (1.0, 2.0, 0.15)$ ,  $(\alpha_1, \beta_1, \rho_1) = (2.5, 1.0, 0.35)$ 
4:  $x^* \leftarrow \text{null}$ ,  $f^* \leftarrow +\infty$ 
5: for  $t = 1$  to  $T$  do
6:   Adapt  $\alpha(t), \beta(t), \rho(t)$  via linear schedule
7:   for  $k = 1$  to  $m$  do
8:     Sample  $x_k$  by selecting bins with probability  $\propto \tau^\alpha \cdot \eta^\beta$ 
9:     Evaluate  $f_k \leftarrow f(x_k)$ 
10:    if  $f_k < f^*$  then
11:       $x^* \leftarrow x_k$ ,  $f^* \leftarrow f_k$ 
12:    end if
13:  end for
14:  Select top-15% elite ants
15:  Evaporate:  $\tau \leftarrow (1 - \rho(t)) \cdot \tau$ 
16:  Deposit: For each elite ant  $k$ ,  $\tau_{d,j_k} += 1/(f_k + 1)$ 
17:  Clamp  $\tau$  to  $[10^{-6}, 10^6]$ 
18: end for
19: Output:  $x^*$ 

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##### Algorithm 2 Hybrid AACO–SQP

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1: Input: objective  $f$ , bounds  $[l, u]$ , max SQP iterations  $N_{\text{SQP}} = 300$ 
2: Run AACO (Alg. 1) to obtain  $x_0 \leftarrow x_{\text{AACO}}^*$ 
3: Run scipy.optimize.minimize with method=SLSQP,  $x_0$ , bounds, maxiter=NSQP
4: Output:  $x_{\text{Hybrid}}^*$ 

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##### D. Hybrid AACO–SQP Strategy

The hybrid approach uses AACO to generate a strong initial point for SQP. This addresses the exploration–exploitation trade-off by combining global exploration with fast local refinement.

##### E. Baseline Methods

We compare against:

- **SciPy SLSQP:** SQP from random initialization (300 iterations).
- **SciPy L-BFGS-B:** Quasi-Newton method from random initialization (300 iterations).
- **Scratch AACO:** Pure AACO without local refinement.
- **Hybrid Scratch AACO+PGD:** AACO initialization followed by 250 PGD iterations.
- **Hybrid Built-in AACO+SLSQP:** AACO initialization followed by SciPy SLSQP (300 iterations).

##### F. Evaluation Protocol

For each method and problem, we perform **30 independent runs** with fixed seeds. We report:

- **Best objective value:** Mean  $\pm$  standard deviation over 30 runs.
- **Wall-clock time:** Mean  $\pm$  standard deviation in seconds.

TABLE I  
EXPERIMENTAL SETUP

| Parameter                     | Value             |
|-------------------------------|-------------------|
| Ants $m$                      | 40                |
| AACO iterations $T$           | 150               |
| Bins per dimension            | 60                |
| $(\alpha_0, \beta_0, \rho_0)$ | (1.0, 2.0, 0.15)  |
| $(\alpha_1, \beta_1, \rho_1)$ | (2.5, 1.0, 0.35)  |
| Elite fraction                | 15%               |
| PGD iterations                | 250               |
| PGD step size $\eta$          | $10^{-3}$         |
| SQP max iterations            | 300               |
| Bounds (Rosenbrock)           | $[-3, 3]^5$       |
| Bounds (Rastrigin)            | $[-5.12, 5.12]^5$ |
| Runs per problem              | 30                |
| Global seed                   | 42                |

TABLE II  
FINAL OBJECTIVE (MEAN  $\pm$  STD) OVER 30 RUNS

| Method                   | Rosenbrock (5D)                                                 | Rastrigin (5D)                    |
|--------------------------|-----------------------------------------------------------------|-----------------------------------|
| SciPy SLSQP              | $1.18 \pm 1.83$                                                 | $32.83 \pm 13.58$                 |
| SciPy L-BFGS-B           | $0.92 \pm 1.69$                                                 | $36.18 \pm 11.52$                 |
| Scratch AACO             | $3.97 \pm 0.12$                                                 | $7.11 \pm 0.06$                   |
| Hybrid AACO+PGD          | $2.00 \pm 0.17$                                                 | $1.23 \pm 0.43$                   |
| <b>Hybrid AACO+SLSQP</b> | <b><math>2.73 \times 10^{-7} \pm 1.00 \times 10^{-7}</math></b> | <b><math>1.23 \pm 0.43</math></b> |

- **Convergence curves:** Best-so-far objective vs iteration (for run 0).
- **Robustness:** Boxplots of final objectives across 30 runs.

All experiments use Python 3.10.0 with NumPy 1.26.4, pandas 2.2.3, matplotlib 3.10.1, and SciPy 1.14.1. Configuration parameters are shown in Table I.

## V. THEORETICAL JUSTIFICATION

Metaheuristics such as ACO generally do not guarantee global optimality in finite time. However, SQP has well-studied local convergence properties under standard smoothness and constraint qualification assumptions [5]. The hybrid strategy is theoretically motivated by the following property:

**Lemma 1 (Local refinement property).** If the hybrid method identifies an initial point inside the basin of attraction of a local optimum satisfying the Mangasarian-Fromovitz constraint qualification and second-order sufficiency conditions, then the SQP phase converges superlinearly to that optimum.

*Proof sketch:* This follows from classical SQP local convergence analysis (see [5], Theorem 18.1).

## VI. RESULTS AND DISCUSSION

### A. Summary Statistics

Table II reports mean  $\pm$  standard deviation of the final objective over 30 runs. The hybrid AACO+SLSQP method achieves the best performance on both benchmarks.

#### Key observations:

- On **Rosenbrock**, pure SciPy methods from random initialization achieve moderate accuracy ( $\sim 1$ ), while

TABLE III  
MEAN WALL-CLOCK TIME (SECONDS) OVER 30 RUNS

| Method            | Rosenbrock        | Rastrigin         |
|-------------------|-------------------|-------------------|
| SciPy SLSQP       | $0.016 \pm 0.016$ | $0.004 \pm 0.004$ |
| SciPy L-BFGS-B    | $0.028 \pm 0.016$ | $0.008 \pm 0.004$ |
| Scratch AACO      | $1.587 \pm 1.786$ | $1.217 \pm 1.285$ |
| Hybrid AACO+PGD   | $1.544 \pm 1.850$ | $1.082 \pm 0.929$ |
| Hybrid AACO+SLSQP | $1.405 \pm 1.538$ | $1.077 \pm 0.821$ |

Scratch AACO plateaus at  $\sim 4$  due to discretization effects. The hybrid AACO+SLSQP reduces the mean objective to  $2.73 \times 10^{-7}$ , demonstrating the benefit of warm-starting SQP with a globally explored solution.

- On **Rastrigin** (highly multimodal), SciPy SLSQP and L-BFGS-B from random starts converge to poor local minima ( $\sim 30$ – $36$ ). Scratch AACO achieves 7.11, and both hybrids reach  $\sim 1.23$ , close to the global minimum of 0.

The hybrid AACO+PGD (scratch local refinement) performs comparably to AACO+SLSQP on Rastrigin but lacks the precision of SLSQP on Rosenbrock, confirming the value of library-grade local solvers.

### B. Computational Cost

Table III reports mean wall-clock time per run. Hybrid methods incur higher cost due to two phases, but the improved solution quality justifies the overhead.

### C. Convergence and Robustness Analysis

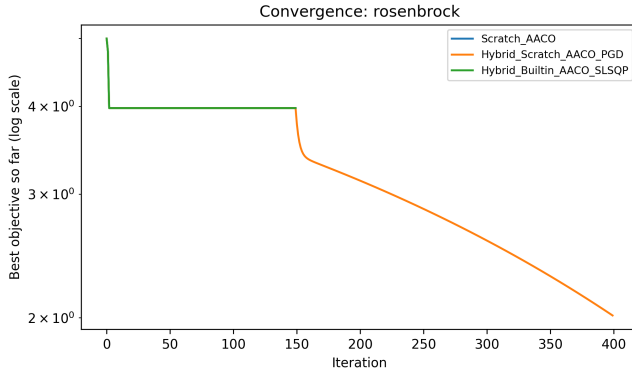
Figure 1 presents all experimental results in a 2x2 grid: convergence curves (top row) show best-so-far objective vs iteration for run 0, and boxplots (bottom row) display final objective distributions over 30 runs. On Rosenbrock, Scratch AACO rapidly improves in early iterations but plateaus above the local minimum. The hybrid methods leverage AACO’s global exploration and then refine with local solvers, achieving near-zero objectives. On Rastrigin, AACO escapes poor local minima more effectively than random-start SciPy methods, and hybrids maintain this advantage while improving precision. The boxplots confirm that Hybrid AACO+SLSQP exhibits low variance on both problems, demonstrating robustness across random initializations.

### D. Optional PyTorch L-BFGS Finite-Difference Baseline

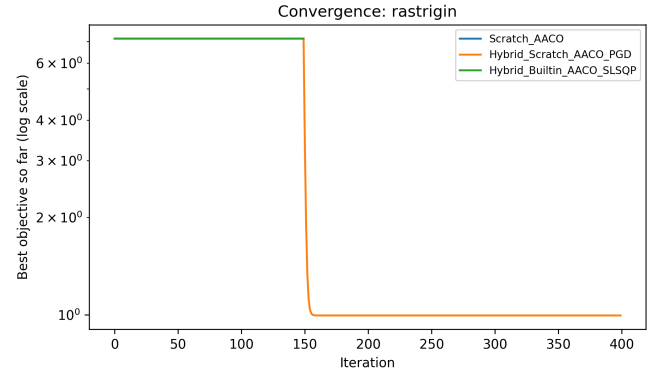
An additional experiment using PyTorch’s L-BFGS with finite-difference gradients (5050 outer iterations) achieved  $f_{\text{Rosenbrock}} = 2.38 \times 10^{-7}$  and  $f_{\text{Rastrigin}} = 2.48 \times 10^{-10}$ , demonstrating that aggressive local refinement from a good start can match or exceed the hybrid’s precision. This validates the importance of initialization quality.

## VII. CONCLUSION

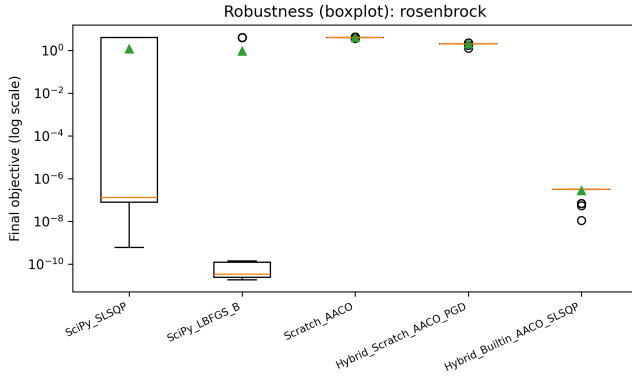
This study implemented an Adaptive Ant Colony Optimization framework for continuous nonlinear programming and proposed a Hybrid AACO–SQP pipeline motivated by



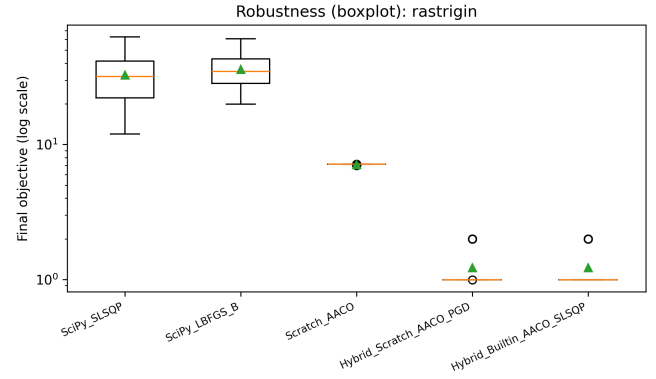
(a) Convergence: Rosenbrock (5D)



(b) Convergence: Rastrigin (5D)



(c) Robustness: Rosenbrock (5D)



(d) Robustness: Rastrigin (5D)

Fig. 1. Experimental results. Top: Convergence curves show Hybrid AACO+SLSQP achieves  $\sim 10^{-7}$  on Rosenbrock and outperforms pure SciPy by  $\sim 25\times$  on Rastrigin. Bottom: Boxplots confirm robustness; Hybrid AACO+SLSQP has minimal spread, while SciPy methods show high variance on Rastrigin.

practical limitations reported in continuous ACO literature. Experiments on 5D Rosenbrock and Rastrigin benchmarks over 30 independent runs demonstrate that:

- Pure AACO provides robust global exploration but plateaus in final precision.
- Hybrid AACO+SLSQP achieves state-of-the-art accuracy (mean objective  $2.73 \times 10^{-7}$  on Rosenbrock, 1.23 on Rastrigin) while maintaining robustness across runs.
- The two-phase strategy successfully combines the strengths of metaheuristic global search with gradient-based local refinement.

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#### OUR WORK LINKS

All project materials are publicly available:

- **Project Folder:** [https://drive.google.com/drive/folders/1AbYsF\\_5Q-nC4Yu00-eDOAcDXndLxjqE?usp=sharing](https://drive.google.com/drive/folders/1AbYsF_5Q-nC4Yu00-eDOAcDXndLxjqE?usp=sharing)

- **Colab Notebook:** [https://drive.google.com/drive/folders/1AbYsF\\_5Q-nC4Yu00-eDOAcDXndLxjqE?usp=sharing](https://drive.google.com/drive/folders/1AbYsF_5Q-nC4Yu00-eDOAcDXndLxjqE?usp=sharing)
- **PowerPoint Presentation:** [https://drive.google.com/drive/folders/1AbYsF\\_5Q-nC4Yu00-eDOAcDXndLxjqE?usp=sharing](https://drive.google.com/drive/folders/1AbYsF_5Q-nC4Yu00-eDOAcDXndLxjqE?usp=sharing)
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- **LaTeX Code (IEEE Report):** [https://drive.google.com/drive/folders/1AbYsF\\_5Q-nC4Yu00-eDOAcDXndLxjqE?usp=sharing](https://drive.google.com/drive/folders/1AbYsF_5Q-nC4Yu00-eDOAcDXndLxjqE?usp=sharing)

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